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The first stars

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Summary. Primordial clouds are likely to be remarkably uniform over stellar mass-scales in the absence of a pre-existing generation of stars. Thermal instability is found to occur during the collapse of a primordial cloud when the H_2 abundance is rising and the H_2 optical depth is of order unity. The e-folding rate for fluctuation growth exceeds the free-fall collapse rate by an order of magnitude. Large density fluctuations of mass-scale $\sim 0.1\,M_\odot$ arise in any collapsing cloud with metallicity $\leqslant 10^{-3}$ of the solar value. Gravitational instability ensures that many of the clumps coagulate to form protostars of masses extending up to the initial Jeans mass when the fluctuations develop, namely $\sim 100\,M_\odot$. The primordial IMF should therefore have spanned the mass range from ~ 0.1 to $\sim 100\,M_\odot$ but may have been dominated by the more massive stars.

1 Introduction

There is general agreement among astrophysicists that the first stars formed from matter almost completely devoid of heavy elements. The lack of significant numbers of disc stars with low metallicity signifies either that the first stars to form in the disc possessed an initial mass function (IMF) that was very different from the present solar neighbourhood IMF, or else that the initial rate of star formation was much reduced per unit mass of gas (*cf.* Audouze & Tinsley 1976). This problem does not arise in the halo, which contains the oldest stars, since gas ejected by stars is not retained in the halo but mostly accumulates in the disc. Consequently, two-component models of our Galaxy are consistent with a Salpeter-like IMF for the halo stars (Ostriker & Thuan 1975), as are also observations of metal-poor halo stars (Caloi, Castellani & Gerola 1983). What can one then deduce about the IMF of the oldest stars?

The requirement to synthesize the heavy elements seen in extreme Population II stars indicates that the preceding stellar generation, Population III, must have contained massive stars. Population III is here taken to refer to an initial stellar population formed with an abundance of heavy elements so low that stellar formation and evolution proceeds as if Z were effectively zero. In practice, this means† $[Z] \leqslant 10^{-3}$ (e.g. Silk 1977a). Detailed studies

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† The square bracket denotes metallicity Z relative to that of the Sun.

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of the abundances of extreme Population II stars with $[Z] \leqslant 10^{-2}$ reveal some similarities and some differences in abundance ratios with Population I stars taken relative to Fe (e.g. Bessell & Norris 1981). Such stars show an enhancement in oxygen [O/Fe] ≈ 0.7 dex relative to the Sun, while [C/Fe] $\approx$ [CNO/Fe] ≈ 0 dex and are not enhanced (Sneden 1979; Barbuy 1983). An oxygen enhancement is indicative of a precursor population that included an excess (relative to the present IMF) of massive stars. Analysis of this and other abundance anomalies in metal-poor stars reinforces this conclusion (Jones 1981).

The enrichment of the intergalactic medium discovered via its diffuse thermal X-ray emission in galaxy clusters must also have been produced by an earlier generation of stars more massive than those presently observed in cluster ellipticals. Again, the observed abundance ratios are about one-third of solar (Mushotzky *et al.* 1978) and indicative of a stellar origin for the intergalactic enrichment. The amount of enriched intergalactic matter is so great ($\sim 10^8 M_\odot$ of Fe per luminous galaxy) and current mass loss and supernova rates are so low (amounting to $\sim 10^{-3} M_\odot\,\mathrm{yr}^{-1}$ per luminous galaxy) that some, if not all, of the enrichment of the intergalactic medium is likely to have originated in ejecta from Population III, where one is not constrained by the observed stellar luminosity and inferred stellar mass loss rates at the present epoch. The discovery that oxygen is enhanced relative to iron ([O/Fe] ≈ 0.7 dex) in the Virgo cluster gas (Canizares *et al.* 1982) is suggestive of an origin for the enriched intergalactic gas in stars similar to those responsible for extreme Population II abundances. The difference in [Fe/H] between the halo and the intergalactic gas presumably reflects the continuous history of star formation in our Galaxy, while the primordial enriched stellar debris accumulates in the intergalactic medium, effectively undiluted by subsequent, much reduced mass-loss from galaxies.

In summary, observational evidence suggests that the mass range of the first stars extended over 10–$100 M_\odot$ and perhaps up to several hundred $M_\odot$, since only such stars are capable of accounting for the nucleosynthetic evidence. In addition, the discovery (Bessel & Norris 1981) of a halo star with [Fe/H] ≈ -4.4 dex indicates that the IMF of the first stars extended to below $1 M_\odot$. Perhaps their IMF was not too different from that of stars forming at the present epoch, apart from the enhanced contribution by massive stars.

It is only when one attempts to simultaneously account for Population III and the dark mass which is responsible for the high M/L ratios in galaxy haloes and clusters that one has to resort to a bimodal IMF, in which the massive stars that produce a minimal level of enrichment are supplemented either by a vast population of 'Jupiters' or by a massive object population that has efficiently hid itself in the form of black holes (e.g. Rees & Kashlinsky 1981). We shall try to avoid such controversies, motivated in part by the possibility that the dark mass may well be non-stellar and non-baryonic (Schramm & Steigman 1981). Restricting our attention to the more narrowly defined Population III problem, we ask: what can be inferred about the mass range of the first stars, the progenitors of Population II?

Our starting point is with primordial clouds in the mass-range 10^6–$10^9 M_\odot$. This mass-range is predicted for pregalactic clouds in different scenarios based on whether adiabatic (Sunyaev & Zel'dovich 1972) or isothermal (Peebles & Dicke 1970; Doroshkevich, Zel'dovich & Novikov 1967) fluctuations are taken to represent the initial conditions in the early Universe. These two modes represent the principal perturbation modes of a Friedmann universe that are relevant for galaxy formation theory. The mass-scale in the adiabatic theory is determined by the minimum fragment size during pancake collapse. These fragments are gravitationally unstable and are not expected to have any substructure since the initial adiabatic fluctuation spectrum had all primordial substructure erased by photon diffusion and viscosity prior to pancake collapse. Within the isothermal theory, the first scales to collapse correspond to that of the minimum Jeans mass ($\approx 10^6 M_\odot$) after matter-radiation decoupling at

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redshift $z \approx 1000$. Substructure could be present, but smaller scales are stable against gravitational instability. Hence it is likely in the isothermal theory that collapse first occurs on scales greater than $10^6 M_\odot$, and despite uncertainty in the form of the primordial fluctuation spectrum, substructure is again likely to be negligible. An independent indication of the first mass-scale to collapse in a hierarchical clustering model is also obtained from the galaxy correlation function: the scale where non-linearity first occurs is in the range 10^8–$10^9 M_\odot$ if the observed correlations grow from the decoupling epoch (e.g. Silk & Norman 1981).

These primordial clouds appear to differ in two distinct ways from conventional star-forming molecular clouds in our vicinity. They lack any significant degree of enrichment in heavy elements and, since there are no pre-existing embedded stars, the primordial clouds are relatively smooth and uniform. Once stars form at an appreciable rate, it is likely that stellar ejecta will play an important role in stirring up and energizing molecular clouds.

Earlier studies of the fragmentation of primordial clouds have argued that it is the lack of heavy elements which may be responsible for the predominance of massive star formation (Silk 1977a; Yoshi & Sabano 1980). This result has lacked conviction, however, since while the temperature of primordial clouds is elevated due to lack of heavy element coolants, the optical depth is simultaneously reduced. Consequently one has to evolve the cloud to higher densities than for a conventional cloud in order to find the point at which the Jeans mass attains its minimum value, often taken to be the relevant scale for star formation. Collapse is nearly isothermal only while the cloud is optically thin to coolant radiation, when the Jeans mass decreases, to eventually rise when the cloud becomes opaque and the temperature begins to increase. In fact, the minimum Jeans mass attained in spherical collapse can be shown to be approximately proportional to $T^{1/4}$ (independently of density and essentially independently of heavy element mass fraction Z) (Rees 1976; Silk 1977a). Of course, the temperature T implicitly depends on Z, but varies by at most a factor 10^3 between primordial and solar abundances. The opacity-limited fragmentation theory was found to yield values of $M^{\min} \approx 0.3 M_\odot$ for primordial clouds (Hoyle 1953; Silk 1977a).

It has been argued that heat input from dissipation of the kinetic energy of collapse can raise $M_J^{\min}$ (Silk 1977a). Of course, this also happens in conventional star formation and there remains little to uniquely distinguish primordial star formation. A more cogent reason for increasing the minimum fragment mass is that for fragments to survive, they should at least have become non-linear density fluctuations before the effects of opacity increase the Jeans mass and inhibit gravitational instability. Gravitational instability in a collapsing cloud yields a secular growth rate, and the final outcome is critically dependent on the initial fluctuation spectrum. Let δ denote the initial fluctuation amplitude. One now finds that the minimum fragment mass is increased over $M_J^{\min}$ by a factor δ^{-1} in spheroidal collapse or $\delta^{-1/2}$ in oblate collapse (Silk 1982). While for conventional star formation, fluctuations are likely to be large on all scales of interest due to the presence and interaction of pre-existing embedded or nearby stars, this is not necessarily the case for the primordial clouds. Thus the clue to the difference between the primordial and present IMF may lie in understanding the origin of the initial fluctuations on stellar mass-scales and the growth of substructure in primordial clouds that are unenriched in heavy elements.

In an interesting series of papers, Sabano & Yoshii (1977) and Yoshii & Sabano (1979, 1980) have explored the role of thermal instability in collapsing gas clouds. They conclude that perturbations of mass $\approx 10 M_\odot$ can grow rapidly. Unfortunately, their calculations neglect opacity effects, which invalidate their results for enriched clouds (Silk 1980; de Jong, Dalgarno & Boland 1980). Moreover, Palla, Stahler & Salpeter (1982) have found that the previously neglected three-body process for H_2 formation can efficiently convert all the atomic hydrogen to H_2 prior to dissociation, in contrast to the peak H_2 fraction of $\approx 10^{-3}$

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that was characteristic of earlier work. This clearly makes it especially important to consider the role of H_2 opacity in evaluating the importance of thermal instability.

In the following sections a treatment of thermal instability in a uniformly collapsing primordial cloud is presented, incorporating the effects of line opacity. The unperturbed model is summarized in Section 2. Simplified forms are adopted for the generalized cooling function that allows an analytic discussion of the dispersion relation (Section 3). A final section summarizes the implications for primordial star formation.

2 Cloud collapse, H_2 chemistry and cooling

Consider an initially uniform collapsing cloud that undergoes spherical collapse. A general form of the energy equation is

$$\frac{du}{dt} - \frac{p}{\rho^2}\frac{d\rho}{dt} = -L. \tag{1}$$

Here ρ is the gas density, p the thermal pressure, u the internal energy per gram, and $L(\rho, T, f)$ is the generalized energy loss function, equal to the net cooling minus heating rates per gram. If x is the H_2 dissociation energy and f the nucleonic fraction in H_2 $(= 2n_{H_2}/n)$, we have

$$u = \frac{3}{2}RT\left[1 - f + \frac{1}{3(\gamma - 1)}\right] - \frac{1}{2}N_A x f \tag{2}$$

where N_A is Avogadro's number, $R\ (= kN_A)$ is the universal gas constant and γ is the effective adiabatic index for the H_2 molecular fraction. If only rotational degrees of freedom are excited, $\gamma = 7/5$; if vibrational degrees of freedom are also excited, $\gamma = 9/7$ (as is the case for $T > 1000$ K). We assume γ is constant, but are not constrained to adopt a specific value.

The remaining equations are straightforward: the equations of state

$$p = (1 - f/2)\rho RT, \tag{3}$$

continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \tag{4}$$

momentum

$$\frac{du}{dt} = -\nabla p/\rho - \nabla \phi \tag{5}$$

and molecular equilibrium

$$df/dt = F, \tag{6}$$

where $F(\rho, T, f)$ is twice the net H_2 formation rate per nucleon minus the H_2 destruction rate per nucleon. Also u denotes the velocity field and ϕ the gravitational potential. It is evident that most of the physics is hidden in the functions L and F, which will now be discussed.

Consider first the limit where the cloud is optically thin to photons emitted by the H_2 rotationally excited levels. One can approximate the cooling rate by the following expression (Hattori, Nakano & Hayashi 1969).

$$\Lambda_{H_2}^{thin} = n_{H_2}(g/g_0)E_{20}g(T)(1 + \zeta)^{-1}\exp(-E_{20}/kT)\ \text{erg cm}^{-3}\,\text{s}^{-1}. \tag{7}$$

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Here A_{20} is the radiative rate from rotational levels $J = 2$ to $J = 0$, $E_{20} = 6\,hB$ (where B is the rotational constant, $hB/k = 85.2$ K) is the energy of the corresponding photons, $\zeta = A_{20}/n\sigma\langle v\rangle$ is the ratio of radiative to collisional de-excitation rates from level 2, and $g(T)$ includes the contribution of molecules in levels $J > 2$ to the cooling. Since collisional rates generally exceed radiative rates at the high densities of interest (see below), we assume a Boltzmann population in each level. An analytic expression for $g(T)$ is now derivable if we adopt

$$A_{J,J-2} = A_{20}(5/162)\,J(J-1)\,(2J-1)^4\,(2J+1)^{-1}\,\mathrm{s}^{-1} \tag{8}$$

and assume $kT \gg hB$. With

$$E_{J,J-2} = 2(2J-1)\,hB, \qquad g_J = 2J+1 \tag{9}$$

we obtain

$$g(T) \approx \frac{\Sigma\, J^7 \exp(-J^2 E_{20})}{\Sigma\, J \exp(-J^2 E_{20})} \tag{10}$$

where the sums are from $J > 2$.

The dominant contribution to $g(T)$ arises from level

$$J_{\max} \approx (7/2\,kT/hB)^{1/2}. \tag{11}$$

Consequently, we can write:

$$g(T) \approx 0.1\,(kT/hB)^3. \tag{12}$$

This provides a reasonable fit to the numerical integration performed by Hattori *et al.* (1969). More accurate formulations of H_2 cooling have been given in later papers (e.g. Hollenbach & McKee 1979), but the above analytic interpolation provides an adequate representation for our purpose.

In view of the cloud collapse, it is appropriate to adopt a large-scale velocity gradient model for molecular line radiative transfer (Goldreich & Kwan 1974; de Jong, Chu & Dalgarno 1975). The optical depth for photons emitted in a $J \to J-2$ transition is

$$\tau_{J,J-2} = n(R/v)\,c(g_J/g_{J-2})\,(\lambda_{J,J-2}/8\pi c)\,A_{J,J-2}, \tag{13}$$

where R is the cloud diameter. We evaluate the optical depth for photons from level $J_{\max}$. Note that the velocity gradient across the cloud, if assumed due to the gravitational collapse, can be written

$$v/R = (\pi G\rho/6)^{1/2} = 10^{0.72} n_6^{1/2}\,\mathrm{km\,s^{-1}\,pc^{-1}}, \tag{14}$$

where $n_6 = n/10^6\,\mathrm{cm}^{-3}$.

The expression for $\tau_{\max}$ ($\equiv \tau_{J_{\max}, J_{\max}-2}$) is then

$$\tau_{\max} = 10^{-0.35} n_6^{1/2} fkT/hB. \tag{15}$$

This represents the optical depth to those photons which are most important for cooling at temperature T. A generalized cooling rate, approximately incorporating the effects of line opacity, can now be written in the form

$$\Lambda = \Lambda^{\mathrm{thin}}\,\frac{1-\exp(-\tau_{\max})}{\tau_{\max}}. \tag{16}$$

This reduces to Λ_{thin} at small $\tau_{\max}$, and yields $\Lambda_{\mathrm{thick}} \propto (v/R)\,(kT/hB)^2$.

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We henceforth adopt for the cooling rate per gram:

$$\Lambda/\rho = A_1 f T^{\epsilon} \rho^{\eta} [1 - \exp(-\tau)]\tau^{-1}; \quad \epsilon = 3, \eta = 0 \tag{17}$$

and

$$\tau = \tau_0 f T n^{1/2} \tag{18}$$

with $A_1 = 10^{-6.7}$ and $\tau_0 = 10^{-5.3}$.

We are primarily interested here in densities such that collisional de-excitation rates exceed radiative decay rates, namely

$$n > A_{J_{\max}, J_{\max}-2}/\langle \sigma v \rangle = 10^{0.6} (kT/hB)^{5/2} \text{ cm}^{-3}. \tag{19}$$

Consideration of lower densities is easily accomplished by setting $\eta = 1$ and taking $\epsilon \approx 2$.

To finally obtain the heat-loss function L, we incorporate a heating term: assume for illustrative purposes that a fraction ϕ of the gravitational energy in bulk motion is thermalized, whence

$$L = \Lambda/\rho - A_2 n^{1/2} T \text{ erg } g^{-1} \text{ s}^{-1}, \tag{20}$$

where $A_2 = 10^{-7.1}\phi$. If the heating and cooling time-scales are assumed to be short compared to the collapse time, then we may set $L = 0$ to yield

$$T = (A_2/A_1)^{1/2} n^{1/4} f^{-1/2} \tau^{1/2} [1 - \exp(-\tau)]^{-1/2} = 10^{2.45} n_6^{1/4} (f/10^{-3})^{-1/2} \text{ K}$$

for

$$\tau \ll 1.$$

Next we consider the formation and destruction of the principal coolant H_2 in order to obtain an expression for F. At low density, H_2 is formed via H^- production (Hirasawa 1969), with the initial electron density being its residual post-recombination value in the isothermal model, and likely to be higher in the adiabatic fragmentation model. The electrons are destroyed by radiative recombination

$$H^+ + e^- \rightarrow H + h\nu, \qquad k_1 = 1.88 \times 10^{-10} T^{-0.64} \text{ cm}^3 \text{ s}^{-1}, \tag{21}$$

and used to make, first H^- by

$$H + e^- \rightarrow H^- + h\nu, \qquad k_2 = 1.83 \times 10^{-18} T \text{ cm}^3 \text{ s}^{-1}, \tag{22}$$

followed immediately by:

$$H + H^- \rightarrow H_2 + e^-, \qquad k_3 = 1.35 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}, \tag{23}$$

to form H_2. The equilibrium H_2 abundance, prior to the onset of dissociation, is evidently:

$$f = k_2/k_1 = 10^{-3.2} T_3^{1.6} \tag{24}$$

where $T_3 = T/1000$ K.

Now an additional channel for forming H_2 is via the previously neglected three-body process (Palla *et al.* 1982),

$$3H \rightarrow H_2 + H, \qquad k_4 = 5.5 \times 10^{-29} T^{-1} \text{ cm}^6 \text{ s}^{-1}, \tag{25}$$

the rate for which exceeds the H^- mechanism rate if

$$k_4 n^2 > k_2 n_e. \tag{26}$$

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To estimate n_e, we equate rate k_1 with the initial collapse time $t_{ff} = (3\pi/32G\rho)^{1/2}$ of the cloud, yielding:

$$n_e = (k_1 t_{ff})^{-1} = 10^{-3.6} n^{1/2} T_3^{0.6}. \quad (27)$$

It follows that the three-body process dominates H_2 formation at densities n exceeding

$$n_{cr} = 10^9 T_3^{1.7} \text{cm}^{-3}. \quad (28)$$

At such densities, the H_2 fraction approaches unity until dissociation sets in via

$$H_2 + H \rightarrow 3H, \quad k_5 = 6.5 \times 10^{-7} T^{-1/2} \exp(-T_0/T) \quad (29)$$

with $T_0 \equiv 51\,800$ K. We also have $2H_2 \rightarrow 2H + H_2$ at rates $k_4/8$ and $k_5/8$ respectively. The H_2 abundance at high density ($> n_{cr}$) is obtained from:

If $f \ll 1$,

$$k_4(1-f)^2 n = k_5 f. \quad (30)$$

$$f \approx 10^{22.5\, T_3^{-1} - 17.6} n_6 T_3^{-1/2} \quad (31)$$

and if $1 - f \ll 1$,

$$1 - f \approx 10^{8.8 - 11.25\, T_3^{-1/2}} T_3^{1/4} n_6^{-1/2}. \quad (32)$$

We now have for the generalized H_2 formation function,

$$F = 2n_e(k_2 - k_1 f) + 2n(1 - 15f/16)\{k_4(1-f)^2 n - k_5 f\}. \quad (33)$$

3 Thermal instability

We are primarily interested in effects on a short time-scale compared to the collapse time. Thus to analyse the cloud thermal stability with respect to small density or temperature perturbations, it is adequate to perturb the cloud about a stationary state, and neglect the gravitational force term. We consider a cloud which is in thermal and chemical equilibrium, so that

$$L(\rho, T, f) = F(\rho, T, f) = 0. \quad (34)$$

Equations (1)–(6) are subjected to an infinitesimal perturbation of the form

$$T = T_1 \exp(ikx + \omega t) \text{ etc } \ldots$$

where the subscript 1 denotes a perturbed variable. The isobaric condition $p_1 = 0$ is adopted, since we are primarily interested in thermally unstable modes.

In Appendix A, a dispersion relation is derived of the form

$$A\omega^2 + B\omega + C = 0. \quad (35)$$

Since $A > 0$, we infer that a sufficient condition for instability is

$$C \equiv [\rho L_\rho - TL_T][F_f + 1/2\rho F_\rho (1-f)^{-1}] + [L_f + (1/2)\rho L_\rho (1-f)^{-1}][TF_T - \rho F_\rho] < 0, \quad (36)$$

where subscripts f, ρ, T denote partial derivatives of $L(f, \rho, T)$ and $F(f, \rho, T)$. In fact, one can readily show that this inequality is also a necessary condition for $Re(\omega) > 0$. This criterion was previously derived by Sabano & Yoshi (1977). The dispersion relation (35)

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reduces to the usual thermal instability criterion (Field 1965) if f = constant, namely $TL_T - \rho L_\rho < 0$. To study the sign of C, we now explicitly evaluate the derivatives of L and F. Define a cooling time-scale

$$t_c = \frac{3}{2}\frac{a^2}{fA_1 T^3}, \tag{37}$$

and let

$$H = d/d\tau \ln [\tau/1 - \exp(-\tau)] = \frac{1-(1+\tau)\exp(-\tau)}{1-\exp(-\tau)}, \tag{38}$$

where

$$0 \leqslant H \leqslant 1, \quad H = 1\ (\tau \to \infty), \quad H = 1/2\tau\ (\tau \ll 1).$$

When the three-body H_2 formation process and H_2 dissociation occur, we also define an H_2^- formation time-scale by

$$t_f^{-1} = \frac{2k_4}{f}(1-f)^2(1-15f/16)n^2. \tag{39}$$

Thermal instability occurs only when three-body formation of H_2 plays an important role at $n \geqslant n_{cr}$ and at optical depths of order unity (Appendix B). The necessary condition is $C < 0$, or from equation (B4)

$$\left(\frac{3}{2}+\frac{T_0}{T}\right)(1-H) < \frac{1+f/2}{1-f}\left[\frac{5-H}{2}+\frac{f}{2}\frac{(3/2)+(T_0/T)}{1+f/2}\right]. \tag{40}$$

Evidently we require the optical depth parameter $H < 1$ (and also $f < 1$) for instability. This means that only at H_2 optical depths $\tau \leqslant 1$ can we expect instability. To quantify this, we combine equations (18) and (20) to yield (eliminating n)

$$f = 10^{-2.1}\phi^{1/2}T_3^{-3/2}\tau^{1/2},$$

where $T_3 \equiv T/1000\,\mathrm{K}$. Hence since $T_3 \approx 1$ and $\phi \leqslant 1$, we can safely assume $f \ll 1$ and $\tau \leqslant 1$ in the region of instability. Equation (35) now simplifies to:

$$\omega^2 + \omega[t_f^{-1} + (3/10)t_c^{-1}(5-H)] - \frac{3}{2t_c t_f}\left[\frac{T_0}{T}(1-H) - 1 - H\right] = 0. \tag{41}$$

The instability condition (46) reduces to:

$$(1+H)/(1-H) < T_0/T \approx 50.$$

This is equivalent to $H < 0.96$ or $\tau \leqslant 5$.

For $\tau \lesssim 1$, $H \approx \tau/2$, and the maximum growth rate obtained by solving (41) is:

$$\omega_{max} \approx 25\, t_f^{-1} \quad (\text{for } t_c \ll t_f),$$
$$\approx 10\, t_c^{-1} \quad (\text{for } t_f \ll t_c),$$
$$\approx 10\,(t_c t_f)^{-1/2} \quad (\text{for } t_f \sim t_c).$$

We are interested in a growth rate that is rapid compared with the free-fall collapse time

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t_{ff}. The relevant time-scales are:

$$t_{ff} = \phi t_c = (3\pi/32G\rho)^{1/2} = 10^{10.14} n_{10}^{-1/2}\,\mathrm{s}, \tag{42}$$

and

$$t_f = f/2k_4 n^2 = 10^{11.0} f T_3\, n_{10}^{-2}\mathrm{s}. \tag{43}$$

Also from (20), if $\tau \lesssim 1$,

$$f = 3/2\, a^2 \phi (A_1 T^3 t_{ff})^{-1} = 10^{-1.49} T_3^{-2} n_{10}^{1/2} \phi,$$

whence

$$t_f/t_{ff} = 10^{-0.67} \phi (T_3 n_{10})^{-1}$$

and

$$\tau = 10^{1.17} \phi n_{10} T_3^{-1}.$$

We can solve (41) after substituting these values for the coefficients. The resulting expression depends on τ and ϕ. It suffices to make the following comments about the inferred growth rate. First, ω is a monotonically decreasing function of τ for $\tau \gtrsim 1$. Equation (47) can be rewritten

$$\frac{t_{ff}}{t_f} = \frac{\Omega\{\Omega + 3/2\,\phi(1-H/5)\}}{3/2\,\phi\,\{T_0/T(1-H) - 1 - H\} - \Omega}$$

where $\Omega = \omega t_{ff}$.

The maximum value of ω can now be expressed as

$$\omega_{\max} t_{ff} = 3/2\,\phi\,\{(T_0/T)(1-H) - 1 - H\}. \tag{44}$$

Now τ cannot be too low otherwise $n < n_{\mathrm{cr}}$, the condition for three-body formation and dissociation to dominate being expressible as

$$\tau > 10^{0.17} \phi T_3^{0.7}. \tag{45}$$

For $\tau \ll 1$, $\omega_{\max}$ becomes insensitive to τ, attaining the limit

$$\omega_{\max} t_{ff} = 75\,\phi\,(1 - \tau/2)\, T_3^{-1}. \tag{46}$$

Moreover, we cannot allow $\tau \gg 1$, otherwise the factor $1 - H$ in (44) severely depresses the maximum growth rate. Hence (45) and (46) imply

$$\omega_{\max} t_{ff} \lesssim 25\, T_3^{-1.7} \tag{47}$$

with equality at $\tau = 1$. Note that this result is independent of ϕ and n: also $T_3 \approx 1$ at $\tau \approx 1$ and depends only logarithmically on other parameters. In fact (18) and (34) yield

$$T = \frac{1434}{1 - 0.03 \ln (T_3^{\frac{2}{3}} \tau^{1/2} \phi^{-3/2})}\,\mathrm{K}. \tag{48}$$

Evidently, the instability is operative with $\omega_{\max} t_{ff} \gtrsim 10$ over a narrow range of density, namely $n = 10^9$–$10^{10}\,\mathrm{cm}^{-3}$, where moreover we require $0.1 \lesssim \phi \lesssim 1$. The optical depth is near unity:

$$1 \gtrsim \tau \gtrsim 10^{0.17} \phi\, T_3^{0.7},$$

the cloud temperature $T \approx 1500\,\mathrm{K}$, and the molecular hydrogen fraction $f = 10^{-1.5}$ to 10^{-3}.

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The instability may suffice, however, to fuel primordial star formation, as will now be argued.

4 Primordial star formation

The length scale that corresponds to the most rapidly growing mode is:

$$l_{\rm th} \approx \pi a^2 \omega_{\rm max}^{-1} \approx 0.1\, l_{\rm J},$$

where $l_{\rm J}$ denotes the Jeans length evaluated at $\tau(H_2) = 1$. The mass associated with this wavelength is:

$$M_{\rm th} \approx \pi \rho l_{\rm th}^3/6 = 10^{-1.14} T_3^{3/2} n_{10}^{-1/2} M_\odot$$

whereas the Jeans mass at this stage amounts to $\approx 100\, M_\odot$.

Our interpretation of the thermal instability is the following. The growth rate of mass-scales $M_{\rm th}$ is sufficiently rapid, occurring with an e-folding time $\lesssim 0.1\, t_{\rm ff}$, that it is likely that large density fluctuations will develop over this scale at a density $n = 10^9$–$10^{10}\,{\rm cm}^{-3}$. Our analysis is valid only in the linear regime, but it suggests that fluctuations with density and temperature contrast of order unity will form in pressure equilibrium. We can only speculate on their fate, but it seems clear that as the Jeans mass decreases in the continuing collapse, smaller and smaller aggregations of clumps of mass $M_{\rm th} \approx 0.1\, M_\odot$ become bound. These collections of clumps are destined to coalesce into individual protostars. A nearly flat mass spectrum results, which extends over ~ 0.1 to $\sim 100\, M_\odot$. Identification of this with the IMF of the first stars suggests that the primordial IMF was enhanced in massive stars of $\lesssim 100\, M_\odot$ relative to the Salpeter IMF, but extended down to $\sim 0.1\, M_\odot$. Moreover this conclusion is exclusive to clouds with metallicity $\lesssim 10^{-3}$ that of the Sun. Once heavy element cooling becomes important, the thermal instability no longer operates. Presumably star formation then operates by processes similar to those occurring in the present interstellar medium.

How secure are these conclusions, based on somewhat crude analytical arguments? From numerical hydrodynamic computations of three-dimensional collapse and fragmentation, we know that the initial Jeans mass is an important parameter in determining the final outcome (Larson 1978; Bodenheimer, Tohline & Black 1980). In primordial clouds, we have found that fluctuations develop spontaneously when the H_2 abundance rapidly increases at $n = 10^9$–$10^{10}\,{\rm cm}^{-3}$, and the Jeans mass at this instant is the most likely measure of the maximum mass-scales to eventually become bound as the collapse proceeds. It seems plausible that the minimum scale which is bound at the onset of collapse provides a measure of the characteristic masses of the final fragments. Unfortunately, the issue of whether further subfragmentation occurs lies outside the dynamic range of existing three-dimensional simulations of collapsing clouds. Plausibly, however, in primordial clouds, the rapidly increasing opacity may mean that subfragmentation is unlikely to play a major role. For example, new subfragments will not have sufficient time to develop and become isolated blobs because of the inhibition of their growth by thermal pressure gradients (Tohline 1980).

Another notable difference arises with rotation. In conventional clouds, magnetic braking enables clouds to overcome the angular momentum barrier to collapse and fragmentation (Mouschovias 1981). Primordial clouds are unlikely to possess any significant magnetic field, and the initial rotation is probably low. Small, isolated stellar systems, such as globular clusters and dwarf spheroidal galaxies, which may exemplify the fossilized remains of primordial clouds, show little rotation. Moreover, theoretical simulations of tidal torques

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between newly formed spheroidal condensations in an expanding universe indicate that if v_{rot} is the peak rotational velocity and σ is the three-dimensional velocity dispersion, the rotational parameter (v_{rot}/σ) at the onset of collapse is ≈ 0.04 (Efstathiou & Jones 1979). For rotational forces to impede collapse, the mean cloud density must increase by a factor $(\sigma/v_{rot})^6$. Primordial clouds that collapse at redshift z will have densities $n \approx 10^2\,\Omega\,[(1+z)/100]^3$ at the onset of collapse, and it is therefore somewhat marginal whether thermal instability is initiated before rotational forces dominate their evolution.

Perhaps the most important aspect of a large initial rotation would be the development of a highly aspherical collapse and formation of a disc. However, our most important parameter, namely H_2 optical depth, depends only on local values of density, temperature and velocity gradient. Thus our conclusions about thermal instability are largely independent of geometry, and should only be strengthened if free-fall collapse is impeded. Once the thermal instability has developed, the gravitational instability scale again depends principally on local quantities. Rotational suppression of the gravitational instability (Goldreich & Lynden Bell 1965) only occurs for sufficiently idealized equilibrium conditions that we again feel confident in arguing that our scenario should be largely unaffected. Indeed, three-dimensional numerical simulations of rotating, collapsing clouds demonstrate that rotation does not significantly impede fragmentation (Bodenheimer *et al.* 1980; Boss 1981).

In summary, we have argued that star formation is a fundamentally different process in primordial clouds as opposed to the present-day interstellar medium, primarily because of the lack of any pre-existing stars. This means that primordial clouds should be remarkably uniform over stellar mass-scales at the onset of collapse. In the case of conventional star formation, previous generations of stars guarantee the presence of fluctuations over a wide range of mass-scales. These clouds are gravitationally unstable over stellar mass-scales at much lower densities than primordial clouds because of the strong cooling by heavy elements to below 100 K. The low densities allow feedback of protostellar energy via mass-loss to play an important role in driving fluctuations (Norman & Silk 1980). In non-primordial molecular clouds, the complex array of cooling transitions effectively quenches any thermal instability (de Jong *et al.* 1980). However, in primordial clouds where H_2 cooling predominates, thermal instability generates the fluctuations in density over a time rapid compared to the collapse time, thereby providing the necessary trigger for star formation. Fluctuations of mass-scale $\approx 0.1\,M_\odot$ arise in any collapsing primordial cloud with metallicity $\leqslant 10^{-3}$ of the solar value, when the mean cloud density $n \approx 10^9$–$10^{10}\,\mathrm{cm}^{-3}$ and $T \approx 1500$ K. Gravitational instability is likely to result in the coagulation of these clumps on scales comparable to the Jeans mass at the epoch when the fluctuations arise, namely $\approx 100\,M_\odot$.

The thermal instability depends on a local criterion, and should not be grossly modified by such effects as rotation or anisotropic collapse. It seems that without rather contrived initial conditions, supermassive objects are unlikely to be the first objects to condense out of pregalactic clouds. Rather, the first stellar objects are stars in the conventional mass range of 0.1 to $100\,M_\odot$. An admittedly crude argument suggests that the more massive stars in this mass-range may predominate. Thus pregalactic and protogalactic nucleosynthesis should produce abundance ratios of heavy elements in concordance with what is observed in old halo stars.

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Appendix A: Derivation of the dispersion relation

It is convenient to introduce a new function

$$P = L - 1/2\, F\left[\frac{a^2}{1-f/2}\,\frac{3\gamma-4}{\gamma-1}+N_A x\right] \tag{A1}$$

where $a^2 = dp/d\rho$ is the square of the isothermal sound speed. After eliminating u_1 from the perturbed equations (4) and (5), we are left with three equations for T_1, ρ_1 and f_1. These yield the following dispersion relation for ω:

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$$\begin{vmatrix} P_f & \rho P_\rho - \omega a^2 & TP_T + \frac{3}{2}\frac{\omega a^2}{1-f/2} & 1-f\frac{\gamma-4/3}{\gamma-1} \\ F_f - \omega & \rho F_\rho & TF_T & \\ \frac{1}{2(1-f)} & 1 & 1 & \end{vmatrix} = 0.$$

Subscripts f, T and ρ denote partial derivatives of the functions F, P and L. After some algebra, we obtain an equation of the form:

$$A\omega^2 + B\omega + C = 0 \tag{A2}$$

where

$$A = (1-f/2)^{-1}\,[5/2 - f\,(2\gamma - 5/2)/(\gamma - 1)]\,,$$

$$B = \frac{3}{2}\rho F_\rho (\gamma-1)^{-1}(1-f/2)^{-1}\left[\phi - 4/3 + \frac{f(\gamma-4/3)+1-\gamma}{2(1-f)}\right]$$
$$+ F_f(1-f/2)^{-1}\left[\frac{3}{2}f\frac{\gamma-4/3}{\gamma-1} - \frac{5}{2} + \frac{f}{2}\right]$$
$$TF_T\left[\frac{1}{2(1-f)} + \frac{3}{2}\frac{\gamma-4/3}{\gamma-1}\frac{1}{1-f/2}\right]$$
$$+ \frac{N_A x}{2a^2}(\rho F_\rho - TF_T)$$
$$+ (TL_T - \rho L_\rho)a^{-2}, \tag{A3}$$

and

$$C = \frac{1}{a^2}(\rho L_\rho - TL_T)\;F_f + \frac{\rho F_\rho}{2(1-f)}$$
$$+ \frac{1}{a^2}\left[L_f + \frac{\rho L_\rho}{2(1-f)}\right](TF_T - \rho F_\rho). \tag{A4}$$

Appendix B: Evaluation of the thermal instability criterion

Using (17), (18) and (20), we obtain

$$TL_T = (2-H)\frac{3a^2}{2t_c}, \qquad fL_f = (1-H)\frac{3a^2}{2t_c}, \qquad \rho L_\rho = -\frac{(1+H)}{2}\frac{3a^2}{2t_c}.$$

Next consider F: if the H^- mechanism dominates, so that $n < n_{cr}$, then

$$F = 2\,(k_2/k_1 - f)\,t_{ff}^{-1}, \tag{B1}$$

and so

$$\rho F_\rho = 0, \qquad F_f = -2ft_{ff}^{-1}, \qquad TF_T = 3.3ft_{ff}^{-1}.$$

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Introducing the H_2 formation time-scale t_f defined by (39), we obtain

$$F_f = -\frac{1+f}{1-f}\frac{1}{t_f}, \qquad \rho F_\rho = \frac{f}{t_f}, \qquad TF_T = -\frac{f}{t_f}\left(\frac{1}{2}+\frac{T_0}{T}\right).$$

Hence from equation (36),

$$C = \frac{3}{2t_c}\left\{-\frac{5-H}{2}\left[F_f + \frac{\rho F_\rho}{2(1-f)}\right] + \left[\frac{1-H}{f} - \frac{1+H}{2(1-f)}\right](TF_T - \rho F_\rho)\right\}. \tag{B2}$$

Now for thermal instability, we require $C<0$, whence

$$\frac{3f}{2t_c t_f}\left\{5 - H + 3.3\left[\frac{1-H}{f} - \frac{1+H}{2(1-f)}\right]\right\} < 0 \tag{B3}$$

if

$n < n_{cr}$,

and

$$\frac{3}{2t_c t_f}\left[\frac{5-H}{2}\frac{1+f/2}{1-f} - \left(\frac{3}{2}+\frac{T_0}{T}\right)\left(1 - H - \frac{f}{2}\frac{1+H}{1-f}\right)\right] < 0 \tag{B4}$$

if $n > n_{cr}$. Now $1 \geqslant H \geqslant 0$ and equals unity as $\tau \to \infty$. Evidently if $n < n_{cr}$, there is no thermal instability, since $C>0$.

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